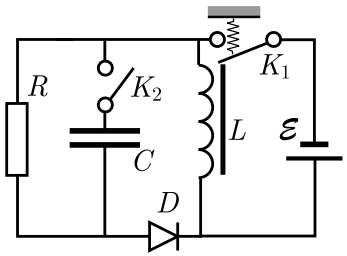
Estonian-Finnish Olympiad 2014

1. DC-DC CONVERTER (8 points) In order to obtain high voltage supply using a battery, the following circuit is used.



An electromagnetic switch K_1 connects a battery of electromotive force $\mathscr{E}$ to an inductor of inductance L: it is closed if there is no current in the inductor (a spring keeps it closed), but if the inductor current reaches a critical value I_0, magnetic field created by the inductor pulls it open. Due to inertia, once the key is open, it takes a certain time τ_K to close again even if the current falls to zero.

For the diode D you may assume that its current is zero for any reverse voltage ($V_D < 0$), and also for any forward voltage smaller than the opening voltage V_0 (i.e. for $0 < V_D < V_0$). For any non-zero forward current, the diode voltage V_D remains equal to V_0.

You may express your answers in terms of L, $\mathscr{E}$, I_0, V_0, and the capacitance C (see figure).

i) *(1 point)* At first, let the key K_2 be open. If the initial inductor current is zero, how long time τ_L will it take to open the key K_1?

ii) *(1 point)* Assuming (here and in what follows) that $L/R \ll \tau_K \ll \tau_L$, plot the inductor current as a function of time t (for $0 \le t < 3\tau_L$).

iii) *(1 point)* What is the maximal voltage $V_{\max}$ on the resistor R?

iv) *(2 points)* Assuming that $V_{\max} \gg V_0$, what is the average power dissipation on the diode?

v) *(2 points)* Now, let the key K_2 be closed, and let us assume simplifyingly that $V_0 = 0$; also, $RC \gg \tau_L$ and $\tau_K > \pi\sqrt{LC}$. Suppose that the circuit has been operated for a very long time. Find the average voltage on the resistor.

vi) *(1 point)* Find the amplitude of voltage variations on the resistor.

2. WASTE PROJECT (8 points) In 2114, Europarliament decided that all radioactive wastes need to be sent to the Sun, so as to avoid contamination of Earth and orbital space. In what follows, you can use the following numerical data: duration of one year $T = 365.25\,\text{days}$, orbital speed on Earth $v_0 = 29.8\,\text{km/s}$, angular diameter of Sun as seen from the Earth $\alpha = 0.5°$, radius of the Earth $R = 6400\,\text{km}$, free fall acceleration at the Earth's surface $g = 9.81\,\text{m/s}^2$.

According to the project, the waste is sent to the Sun using ballistic spaceships: the engine operates only during a short period of time during which the displacement of the spaceship remains much shorter than the radius of Earth. In the Earth's frame of reference, the spaceship obtains a velocity opposite to the orbital velocity of Earth in the Sun's frame of reference. Further, the ship moves along a ballistic trajectory until it hits the Sun. The trajectory is such as to minimize the consumption of fuel.

i) *(1 point)* Sketch the trajectory of the spaceship.

As a first approximation, calculations can be done when neglecting the angular size of the Sun (i.e. by putting $\alpha \approx 0°$); you can use this approach for the next two questions.

ii) *(1.5 points)* How long will it take for the spaceship to travel from the Earth to the Sun?

iii) *(1.5 points)* What is the speed of the spaceship in the Earth's frame of reference when the distance from the Earth is much larger than the Earth's radius, but still much shorter than the distance to the Sun?

iv) *(2.5 points)* Answer the previous question without making the approximation $\alpha \approx 0°$.

v) *(1.5 points)* What is the speed of the spaceship in the Earth's frame of reference when the distance from the Earth is much smaller than the Earth's radius?

3. MAGNETS (6 points) To explore the force between two small magnets, the following experiment is performed. One of the magnets is hanged from a thread with length $l = 1\,\text{m}$. Other magnet is moved slowly closer while keeping the axes of the magnets always on the same horizontal line. At the moment when the distance between the magnets is $d_1 = 4\,\text{cm}$ and the hanged magnet has moved $x_1 = 1\,\text{cm}$ from initial position, balance is lost and the magnets pull together. By making the assumption that the pulling force between the magnets F_m depends on the distance d according to the relation $F_\text{m} \propto d^{-n}$, find the value of the exponent n.

4. SUPERBALLS (5 points) $n+1$ elastic balls are dropped so that they are exactly above each other, with a very small gap between each. Bottom ball has a mass of m_0, the one above has a mass of fm_0, next f^2m_0 and so on, until the topmost ball with mass $f^n m_0$, where $f < 1$. At the moment when the bottom ball touches the ground, all the balls are moving with the speed v.

i) *(1 point)* After the collision between the two bottommost balls, what is the speed v_1 of the second ball from the bottom?

ii) *(3 points)* What is the speed of the topmost ball v_n after all collisions?

iii) *(1 point)* How many times higher would that ball fly compared to the initial drop height h? Take $f = 0.5$ and $n = 10$.

It maybe useful that that sequence $a_0 = 1$, $a_{k+1} = a_k\alpha + \beta$ has a general term $a_n = \alpha^n + \beta\frac{\alpha^n - 1}{\alpha - 1}$, where α and β are constants.

5. PLANCK'S CONSTANT (8 points) In a simplistic model, light emitting diodes can be considered to only pass current when lit, and then they have a constant voltage drop $V_t = \frac{E}{e}$ across them. $E = hf$ is the energy of the light quanta emitted and $e = 1.60 \times 10^{-19}\,\text{C}$ is the elementary charge. Speed of light in vacuum $c = 3.00 \times 10^8\,\text{m/s}$.

You have a assorted light emitting diodes numbered 1–6, each with a $R = 680\,\Omega$ series resistor. From the datasheets it is known for the peak wavelengths of the diodes to be 940 nm, 620 nm, 590 nm, 525 nm, 470 nm, and 450 nm.

i) *(2 points)* Find out which wavelength corresponds to which diode.

ii) *(4 points)* Measure the Planck's constant h that corresponds to our simplistic model. This does not have to correspond to real Planck's constant.

iii) *(2 points)* Estimate the uncertainty.

Equipment: voltage source (batteries) with an unknown voltage, ammeter, assorted light emitting diodes with series resistor. Take care not to short the battery with the ammeter. You may ignore the internal resistances of the batteries and the ammeter.

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6. RUNNING ON ICE (4 points) A boy is running on a large field of ice with velocity $v = 5\,\text{m/s}$ toward the north. The coefficient of friction between his feet and the ice is $\mu = 0.1$. Assume as a simplification that the reaction force between the boy and the ice stays constant (in reality it varies with every push, but the assumption is justified by the fact that the value averaged over one step stays constant).

i) *(2 points)* What is the minimum time necessary for him to change his moving direction to point towards the east so that the final speed is also $v = 5\,\text{m/s}$?

ii) *(2 points)* What is the shape of the optimal trajectory called?

7. SPIN SYSTEM (8 points) Let us consider a system of N independent magnetic dipoles (spins) in a magnetic field B and temperature T. Our goal is to determine some properties of this system by using statistical physics. It is known that the energy of a single spin is $E = \epsilon m$, where $m = \pm\frac{1}{2}$ and $\epsilon = \alpha B$.

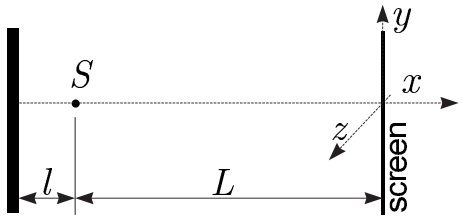
i) *(2 points)* What is the probability $p_\uparrow$ for a spin to be in exited state, i.e. have positive energy?

ii) *(2 points)* What is the average value of the total energy E_s of the spin system as a function of B and T?

iii) *(2 points)* Using high temperature approximation $T \gg \frac{\alpha B m}{k}$, simplify the expression of E_s.

iv) *(2 points)* Using high temperature approximation $T \gg \frac{\alpha B m}{k}$, find the heat capacity C of the spin system.

8. MIRROR INTERFERENCE (5 points) A point source S emits coherent light of wavelength λ isotropically in all directions; thus, the wavefronts are concentric spheres. The waves reflect from a dielectric surface placed at a distance $l = N\lambda$ (where N is a large integer) from the point source, and the interference pattern is observed on a screen which is placed to a distance $L \gg l$ from the point source (see figure).



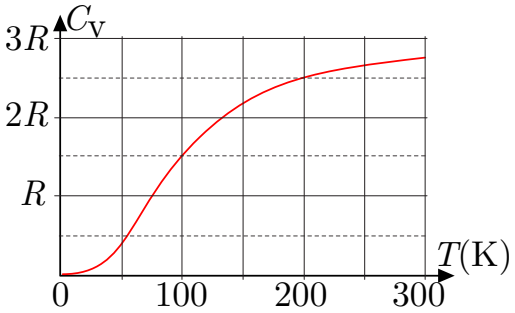
In what follows we use the x, y, and z coordinates as defined in the figure. The screen is parallel to the mirror and lies in the $y-z$-plane.

i) *(2 points)* At which values of the y-coordinate (for $z = 0$) are the interference maxima observed on the screen? You may assume that $y \ll L$.

ii) *(1 point)* Sketch the shape of few smallest-sized interference maxima on the screen (in $y-z$-plane).

iii) *(2 points)* Now the flat screen is replaced with a spherical screen of radius L, centred around the point source. How many interference maxima can be observed?

9. THERMAL ACCELERATION (9 points) Consider a cube of side length $a = 1\,\text{cm}$, made of aluminium (density $\rho = 2.7\,\text{g/cm}^3$, molar mass $M_A = 23\,\text{g/mol}$). The heat capacitance of one mole of aluminium is given as a function of temperature in the graph below. The speed of light $c = 3 \times 10^8\,\text{m/s}$, universal gas constant $R = 8.31\,\text{J/(kg·K)}$. The initial temperature of the cube is $T_0 = 300\,\text{K}$.



i) *(1 point)* What is the total heat energy of such a cube at the initial temperature T_0?

ii) *(3 points)* Now, the cube has 5 faces painted in white (reflects all relevant wavelengths) and one face painted in black (absorbs all these waves). The cube is surrounded by vacuum at a very low temperature (near absolute zero); there is no gravity field. Initially, the cube is at rest; as it cools down due to heat radiation, it starts slowly moving. Estimate its terminal speed v_1.

iii) *(2 points)* At very low temperatures, the heat capacitance of aluminium is proportional to T^3, where T is its temperature. Which functional dependance $f(t)$ describes the temperature as a function of time [$T = A \cdot f(Bt)$, where A and B are constants] for such very low temperatures under the assumptions of the previous question?

iv) *(3 points)* Now, the cube has 5 faces covered with a thermal insulation layer (you may neglect heat transfer through these faces). One face is left uncovered. The cube is surrounded by hydrogen atmosphere at a very low temperature (molar mass of hydrogen molecules $M_H = 2\,\text{g/mol}$). The cube starts cooling down due to heat transfer to the surrounding gas; you may neglect the heat radiation. Initially, the cube is at rest; as it cools down, it starts slowly moving. Estimate the order of magnitude of its terminal speed v_2. Assume that the surrounding gas is sparse, so that the mean free path of the molecules is much larger than a. Assume that $v_2 \ll c_s$ where c_s is the speed sound in the atmosphere surrounding the cube.

10. YOUNG'S MODULUS OF RUBBER (12 points) The linear Hooke's law for a rope made from an elastic material is supposed to held for small relative deformations $\varepsilon = x/L$ (which is also called "strain"), where L is the undeformed length of the rope, and x is the deformation. Once ε becomes too large, the force-deformation relationship $F = kx$ is no longer linear; what is "too large" depends on the material. For very elastic materials which can reach relative deformations considerably large than one, it may happen that the linear Hooke's law with a constant stiffness k fails, but if we take into account the change of the cross-sectional area S of the rope with $k = ES/L$, where E is the Young's modulus of the elastic material, such a non-linear Hooke's law remains valid. In that case we can say that there is still a linear stress-strain relationship $\sigma = E \cdot \varepsilon$, where the stress $\sigma = F/S$.

i) *(7 points)* Measure the relationship between the stress σ and strain ε in a rubber string and plot it.

ii) *(5 points)* From your plot determine the Young's modulus E with its uncertainty, and the maximum strain ε_m until which it applies.

Note: the diameter of the thread is to be measured using the diffraction of laser light.

Equipment: rubber thread, stand, measuring tape, 15 hex nuts with a known mass, a plastic bag for hanging a set of nuts to the thread, a green laser ($\lambda = 532\,\text{nm}$), a screen.

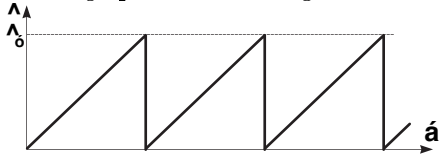
WARNING: AVOID LOOKING INTO A LASER BEAM, THIS MAY DAMAGE YOUR EYES!

1. DC-DC converter

1) (1 pt) From the Kirchoff's voltage law for the loop consisting of L and $\mathcal{E}$, $\mathcal{E} = L\frac{dI}{dt}$, hence $I = \mathcal{E}t/L$. From $I_0 = \mathcal{E}\tau_L/L$ we obtain

$$\tau_L = LI_0/\mathcal{E}.$$

2) (1 pts) Once the current I_0 is reached, the key is opened; the current trough L cannot change instantaneously and therefore is forced to flow through the resistor R. Since the characteristic time of this current loop (consisting of L and R is very short ($L/R \ll \tau_K$), the current decays very fast and becomes essentially equal to zero while the key is still open. Now there is no current through the inductor, so that the key will close again and the process will start repeating from the beginning. As a result we'll have a periodic graph as shown in figure.



3) (1 pt) The voltage through the resistor is maximal when the current is maximal, which happens immediately after the switch is opened; the maximal current is I_0 so that $V_{\max} = RI_0$.

4) (2 pts) Due to $V_{\max} \gg V_0$, we can neglect the effect of the diode; so we have the Kirchoff's voltage law $L\frac{dI}{dt} = RI = R\frac{dq}{dt}$ (here we expressed the current via the charge q which flows through the resistor). Integration over a single cycle (when the inductor current drops from I_0 down to 0) yields $LI_0 = R\Delta q$, hence the charge flown through the resistor (and through the diode) $\Delta q = I_0 L/R$. During that cycle, the diode had a constant voltage V_0, so the electric field performed work $A = V_0\Delta q$ which was released as heat in the diode. So, the average power dissipation

$$P = \frac{A}{\tau_L} = \frac{V_0 I_0 L}{R\tau_L} = \frac{V_0\mathcal{E}}{R}.$$

5) (2 pts) Now, since the characteristic time of the RC-loop is very large, the capacitor maintains its charge (and voltage) during that period of time when the diode is closed. Once the key K_1 opens, the diode will open, and the capacitor is connected to the inductor so that a LC-loop is formed. That loop admits oscillations of period $T_{LC} = 2\pi\sqrt{LC}$ and as the current to the resistor can be neglected (since $RC \gg T_{LC}$), we can neglect the presence of the resistor. During the time interval when the diode is open, we can also neglect the presence of the diode since $V_0 \ll V_{\max}$. Hence, the current i in the LC-loop changes in time sinusoidally, starting from $i = I_0$ and ending when $i = 0$ (then the diode will close disconnecting the LC-loop). During that process, the magnetic energy of the inductor $\frac{1}{2}LI_0^2$ is converted into the electric field energy of the capacitor, which is later released as heat on the resistor. When the stationary regime is achieved, the energy lost by the capacitor during one period (of duration τ_L) as the heat dissipation on the resistor $Q = V_{av}^2\tau_L/R$ must be equal to the energy received from the inductor; so,

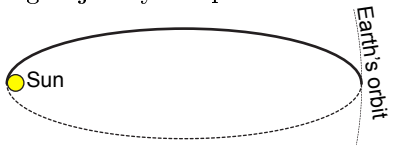
$$\frac{V_{av}^2\tau_L}{R} = \frac{1}{2}LI_0^2 \Rightarrow V_{av} = I_0\sqrt{\frac{LR}{2\tau_L}} = \sqrt{\frac{\mathcal{E}I_0R}{2}}.$$

6) (1 pt) The charge which flows away from the capacitor when the diode is closed can be found as $q_C = \tau_L V_{av}/R$ (owing to $RC \gg \tau_{LC}$, the relative change of the capacitor's voltage is small). Hence, the voltage drop is found as $\Delta V = q_C/C = \tau_L V_{av}/(RC)$. The amplitude is half of the difference between the minimal and maximal values, so that the amplitude

$$U_0 = \frac{\Delta V}{2} = \frac{\tau_L V_{av}}{2RC} = \frac{I_0 L}{2C}\sqrt{\frac{I_0}{2R\mathcal{E}}}.$$

2. Waste project

1) The trajectory is a very elliptical ellipse, perihelion of which is within the Sun. In order to minimize the fuel consumption, the speed near the Earth's orbit needs to be as large as possible (we need to decelerate the ship to bring it to the elliptical orbit), hence the full orbital energy of the ship $E = -\frac{GMm}{2a}$ needs to be as small as possible. Here, M is the mass of the Sun, m — the mass of the space ship, and a — the longer semiaxis. So, a needs to be as large as possible, which means that the perihelion needs to lie at the surface of the Sun, in which case $2a = R_E + r_S$, where R_E is the orbital radius of the Earth and r_S — the radius of the Sun. The resulting trajectory is depicted below.



2) If we neglect the radius of the Sun, the space ship needs to "fall" directly to the Sun, which means that its initial orbital speed must be zero, hence its trajectory is an ellips with longer semiaxis equal to $R_E/2$. According to the Kepler's III law, the period on such an orbit is $2^{3/2}$ times shorter than the Earth's orbital period T. The travel time t is half of the period, so that $t = 2^{-5/2}T \approx 64.6$ days.

3) In the Sun's frame of reference, the speed needs to be zero; hence, in the Earth's frame of reference, it is opposite to the orbital velocity of the Earth and by modulus equal to $v_0 = 29.8\,\mathrm{km/s}$.

4) The speed v_S in the Sun's frame of reference is found from the expression for the total energy,

$$-\frac{GMm}{2a} = -\frac{GMm}{R_E + r_S} = -\frac{GMm}{R_e} + \frac{mv_S^2}{2},$$

hence

$$v_S = \sqrt{\frac{GM}{R_E}\frac{2r_S}{R_E + r_S}}.$$

This expressing can be rewritten by using equality $v_0^2 = \frac{GM}{R_E}$ as

$$v_S = v_0\sqrt{\frac{2r_S}{R_E + r_S}} = v_0\sqrt{2\sin\left(\frac{\alpha}{2}\right)} \approx v_0\sqrt{\alpha}.$$

Numerically this yields $v_S \approx 2.8\,\mathrm{km/s}$; the speed in the Earth's frame of reference $v_E = v_0 - v_S \approx 27.0\,\mathrm{km/s}$.

5) Part of the initial kinetic energy in the Earth's frame of reference goes to the change of the potential energy due to the gravitational pull of the Earth, $\Delta\Pi = \frac{GM_E m}{R} = gmR$ hence $gR + \frac{v_E^2}{2} = \frac{u^2}{2}$. Here, M_E is the Earth's mass and u is the launching speed. So,

$$u = \sqrt{v_E^2 + 2gR} \approx 29.2\,\mathrm{km/s}.$$

3. Magnets

There three forces acting on the hanging magnet: the downwards directed gravity force $m\vec{g}$, the tension force $\vec{T}$, which is directed along the thread, and the horizontal magnetic force $\vec{F}_{\mathrm{m}}$. Since the thread's angle is very small, the modulus of $\vec{T}$ is almost equal to mg, so that its horizontal projection is expressed as $-(x/l)mg$, where l is the length of the thread and $x = 1\,\mathrm{cm}$ (the displacement from the initial position). The net horizontal force $F = F_{\mathrm{m}} - (x/l)mg$. At the equilibrium point $F = 0$. The equilibrium point is stable if a small (virtual) displacement Δx gives rise to a returning force

$$\Delta F = \Delta F_{\mathrm{m}} - \frac{\Delta x}{l}mg$$

which needs to push towards the equilibrium point. Let $F_{\mathrm{m}} = kd^{-n}$, where k is an unknown proportionality coefficient. Then

$$\Delta F_{\mathrm{m}} = F'_{\mathrm{m}}(d)\Delta d = \frac{kn}{d^{n+1}}\Delta x,$$

because $\Delta d = -\Delta x$. Therefore

$$\Delta F = \left(\frac{kn}{d^{n+1}} - \frac{mg}{l}\right)\Delta x.$$

At the limit case of the loss of stability (which is described by this problem) $\Delta F = 0$. Thus we have two equations with two unknowns (n and k):

$$\frac{k}{d^n} - \frac{xmg}{l} = 0, \quad \frac{kn}{d^{n+1}} - \frac{mg}{l} = 0;$$

this can be rewritten as

$$\frac{k}{d^n} = \frac{xmg}{l}, \quad \frac{kn}{d^{n+1}} = \frac{mg}{l}.$$

If we divide the corresponding sides of the two equations we obtain $d/n = x$, hence

$$n = d/x = 4.$$

4. Superballs

1) During the collision with the floor, the bottom-most ball will retain its speed and change the direction of the velocity; its upwards speed $v_0 = v$. Let the k-th ball move up with a velocity v_k; we'll consider the collision between this and $(k+1)$-st ball, which moves down with the velocity v. In the frame of reference of the centre of mass, $u = \frac{v_k - fv}{1+f}$; hence, after the collision the upwards velocity equals to $v_{k+1} = v + 2\frac{v_k - fv}{1+f} = \frac{2}{1+f}v_k + \frac{1-f}{1+f}v$. With $v_0 = v$, we can conclude that $v_1 = \frac{3-f}{1+f}v = \left(\frac{4}{1+f} - 1\right)v$. **2)** One can see that if we apply the recurrent formula repetitively, the result at the n-th step will be $v_n = \left(2\left(\frac{2}{1+f}\right)^n - 1\right)v$.

3) Now we need to relate the speeds to the jumping heights via $v^2 = 2gh_0$ and $v_n^2 = 2gh_n$; hence,

$$h_n/h_0 = \sqrt{v_n/v_0} = \sqrt{2\left(\frac{2}{1+f}\right)^n - 1}.$$

For $f = 0.5$ and $n = 10$ we obtain that the final height will be *ca* 1200 larger than the initial one.

5. Planck's constant

1) When we connect each of the diodes to the battery, we can observe the light of the emitted light; the mapping is as follows: 940 nm — invisible (infrared), 620 nm — red, 590 nm — orange, 525 nm — green; 470 nm — blue; 450 nm — violet.

2) We can measure the current I through the diode (which is also the current through the resistor R), so that the voltage on the diode would

be $V_d = \mathcal{E} - IR$, but we don't know the battery voltage. However, we do know that the diode's voltage equals approximately to the photon's energy E_p expressed in electron volts, $E_p = hc/(\lambda e)$. Since we expect that $IR = \mathcal{E} - V_d$, if we plot IR versus $1/\lambda$, we should obtain a straight line

$$IR = \mathcal{E} - \frac{1}{\lambda}\frac{hc}{e}.$$

We can measure the slope of the straight line $A = hc/e$, which allows us to calculate $h = eA/c$.

3) The major source of the uncertainty is not the instrument uncertainties, but the departure of the real diode data from the simplistic model. Therefore we can try to fit the data points with different straight lines making the slope A as steep as possible (while still keeping a reasonable fit with the data points, and also as flat as possible; the uncertainty of A is found as $\Delta A = \frac{1}{2}(A_{\max} - A_{\min})$, and $\Delta h = h\Delta A/A$.

6. Running on ice

1) During the process, the velocity vector needs to change its direction by 90 degrees. Let us consider the this graphically using the $v_x - v_y$-plane: we need to move from the point A with coordinates $(0, v)$ to a point B with coordinates $(v, 0)$ while having a constant "speed". Indeed, the velocity of a point in the $v_x - v_y$-plane is the acceleration of the body, which has here a constant modulus μg. Obviously, the fastest path is a straight line of length $v\sqrt{2}$, so that $t = v\sqrt{2}/\mu g \approx 7.2\,\mathrm{s}$.

2) Since the direction of the acceleration is constant, the trajectory is the same as for a body in the Earth's field of gravity — a parabola.

7. Spin system

1) According to Boltzmann's distribution, $p_\uparrow = A \cdot e^{-\epsilon m}$, where the constant A can be found from the condition that the probability of having either up or down orientation is one: $A \cdot e^{-\epsilon/2} + A \cdot e^{\epsilon/2} = 1$, hence

$$A = \frac{1}{e^{-\epsilon/2kT} + e^{\epsilon/2kT}} = \frac{1}{2\cosh(\epsilon/2kT)}.$$

Thus,

$$p_\uparrow = \frac{e^{-\epsilon/2kT}}{e^{-\epsilon/2kT} + e^{\epsilon/2kT}}.$$

2) The average energy is the weighted average of up- and down-state energies for a single spin, multiplied by the number of spins:

$$E = \frac{N\epsilon}{2}\frac{e^{-\epsilon/2kT} - e^{\epsilon/2kT}}{e^{-\epsilon/2kT} + e^{\epsilon/2kT}} = -\frac{N\epsilon}{2}\tanh(\epsilon/2kT).$$

3) For small values of the argument of the hyperbolic tangent, the last expression can be approximated as $E \approx -N\epsilon^2/4kT$.

4) According to the definition of the heat capacity, $C = \frac{dE}{dT} = N\epsilon^2/4kT^2$.

8. Mirror interference

1) For a position y, the arriving rays form an angle $\alpha = y/L$ (we use the small-angle approximation; the angle is in radians). Then, the optical path difference between the reflected and direct rays is $\Delta = 2l\cos\alpha \approx 2N\lambda - N\lambda\alpha^2$. Since there is an additional phase shift for the reflected rays at the reflection from optically denser dielectric material, the total phase shift is $\varphi = 2\pi\Delta/\lambda = 4\pi N - \pi(2N\alpha^2 - 1)$. At the maxima, this equals to $2\pi(2N - n)$, where n is an integer. Therefore, the condition for the maxima is written as

$$\alpha = \sqrt{\frac{n+0.5}{N}} \Rightarrow y_n = L\sqrt{\frac{n+0.5}{N}},$$

where $n = 0, 1, \ldots \ll N$.

2) Since the rays of a given order number n form a fixed angle with the x-axis, the maxima form on the screen concentric circles; the pitch between the neighbouring circles becomes smaller as the order number n grows (using the length unit defined by the smallest radius, the radii form a sequence $\sqrt{1} = 1$, $\sqrt{3} \approx 1.73$, $\sqrt{5} \approx 2.23$, etc).

3) Since the reflected rays can reach the screen only within a hemisphere, the phase shift varies between $\varphi_{\max} = 4\pi N + \pi$ and $\varphi_{\min} = \pi$. The number of maxima

$$m = (\varphi_{\max} - \varphi_{\min})/2\pi = 2N.$$

9. Thermal acceleration

1) For the heat energy of one mole of material $dq = C_v dT$. There is no heat energy by $T = 0$, hence $q = \int_0^T C_v dT$. Using the graph we find this as the area under the curve, $q \approx R \cdot 560\,\mathrm{J/K}$. The number of moles $\nu = a^3\rho/M_A \approx 0.117\,\mathrm{mol}$, hence the total heat energy $Q = q\nu \approx 546\,\mathrm{J}$.

2) Each photon of frequency ν radiated by the cube carries away heat energy equal to $E = h\nu$, and carries momentum $p = h/\lambda = h\nu/c = E/c$. If the photon departs at the angle α with respect to the surface normal then the component parallel to the surface normal $p_\parallel = \frac{E}{c}\cos\alpha$. The total momentum given by the photons to the cube equals by modulus to the total momentum carried by the photons; when averaged over all the photons, the perpendicular to the surface normal components cancel out (photons go to all the directions). The average value of the parallel component can be estimated just as $p_\parallel \sim \frac{E}{c}$.

If we want to obtain an exact result, we need to integrate over all the directions while keeping in mind that the light intensity is proportional to $\cos\alpha$. So, the momentum averaged over all the directions $\bar{p}_\parallel = \frac{E}{c}\frac{1}{2\pi}\int\cos^2\alpha d\Omega$, where the solid angle differential $d\Omega = 2\pi\sin\alpha d\alpha$. Therefore, $\bar{p}_\parallel = \frac{E}{c}\int_0^{\pi/2}\sin\alpha\cos^2\alpha d\alpha = \frac{E}{c}\int_0^{\pi/2}\cos^2\alpha d\cos\alpha = \frac{E}{3c}$.

Since the momentum-energy ratio is the same for all the photons, equal to $1/c$, the overall momentum equals to Q/c. Thus, $a^3\rho v \sim Q/c$, hence

$$v \sim \frac{Q}{\rho a^3 c} \approx 0.67\,\mathrm{mm/s}.$$

If we apply the exact factor $\frac{1}{3}$ (obtained above via integration), we end up with $v \approx 0.22\,\mathrm{mm/s}$.

3) The heat balance at very low temperatures can be written as $AT^3 dT = -\sigma S T^4 dt$, where A is a constant, σ is the Stefan-Boltzmann constant, and S — the radiating area. This simplifies to

$$\frac{dT}{T} = -Bt \Rightarrow T = A \cdot e^{-Bt}.$$

4) In the case of a hydrogen atmosphere, the momentum is given to the cube due to the fact that the molecules colliding with the coated faces bounce back with the same speed as the they came, but uncoated face gives away heat energy, and the molecules leave at higher temperature. If we assume that the departing molecules have the same temperature as the cube (which serves us only as an estimate — when particles of different masses collide, only a part of the energy is transferred), then the momentum-to-heat ratio is estimated as $1/v_T$, where $v_T = \sqrt{RT/M_H}$ is the thermal speed of the molecules after the collision with the cube for the motion along the surface normal. So we estimate $a^3\rho v \sim Q/v_T$, hence

$$v \sim \frac{Q}{\rho a^3}\sqrt{\frac{M_H}{RT}} \approx 180\,\mathrm{m/s}.$$

It should be noted that in fact, one should have been careful with such an estimate, because the thermal speed is at the denominator. This will increase the relative contribution of the heat radiated at low temperatures. However, the remaining heat at low temperatures is proportional to T^4, and therefore the contribution of those molecules which collide with the cube at low temperatures to the overall momentum remains still small.

10. Young's modulus of rubber

1) The setup is as follows. The rubber thread is fixed to the stand, and the plastic bag is fixed to the free end of the thread. The hex nuts are added, one-by-one, into the bag, starting with zero and ending with 15. The laser light is directed to the thread and the diffraction pattern is observed on the screen (which is a vertically fixed sheet of graphic paper on a support). The diffraction pattern from a wire is the same as from a single slit (the superposition of the Huygens sources from those two cases gives a full set of sources on a flat wave front, hence the electric fields from those two cases must provide equal and opposite wave amplitudes and equal intensities). So, if we measure on the screen the distance a between such maxima which are separated by n (e.g. $n = 10$) periods of the diffraction pattern then using small-angle-approximation, $n\lambda/d = a/L$, where L is the distance from the thread to the screen and d — the thread diameter. So,

$$d = nL\lambda/a.$$

The strain is calculated by making markings on the thread and measuring the distance b between these in a stretched state,

$$\varepsilon = \frac{b - b_0}{b_0},$$

where b_0 is the length in a non-strained state. The stress is calculated as

$$\sigma = \frac{4Nmg}{\pi d^2},$$

where N is the number of hex nuts in the bag and m — the mass of a single nut. The data are plotted in a graph; linear relationship corresponds to a straight line. The uncertainties are calculated using the rule of relative uncertainties, either using Pythagorean or simple addition, e.g.

$$\Delta\varepsilon = \varepsilon\Delta b\left(\frac{2}{b - b_0} + \frac{1}{b_0}\right),$$

where $\Delta b \approx 0.5\,\mathrm{mm}$ is the length measurement uncertainty. Similarly,

$$\Delta\sigma = 2\sigma\frac{\Delta d}{d}.$$

These uncertainties are marked in the graph as error bars.

2) Using the plot, we need to determine such a value of $\varepsilon = \varepsilon_*$ that for $\varepsilon_1 > \varepsilon_*$, it is impossible to draw a straight line intersecting the error bars of all the data points with $\varepsilon_1 < \varepsilon_*$.